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Abstract— Generative AI in FinTech is a new technology that uses AI to generate images based on predefined lists of Stock Keeping Units (SKUs) using product descriptions provided by users. It leverages an AI-based tool to produce custom product images to accurately sync with individual SKUs. This technology allows FinTech companies to create product images to help promote their ecommerce sites or offline stores, while still adhering to product description specifics that define user intent. With more accurate product images, customers are able to more easily find the product they are looking for. Generative AI in FinTech also helps to reduce the manual effort it takes for businesses to define and recreate product images for customers, saving cost and time. Overall, this technology is a key breakthrough in FinTech's efforts to create seamless customer experiences.

Keywords— Generative AI, FinTech, Generative Imaging, Stock Keeping Units, Product Descriptions

I. INTRODUCTION

Generative AI in FinTech provides an interesting and useful application of artificial intelligence in the financial sector. By developing algorithms to generate images based on predefined lists of Stock Keeping Units (SKUs) using product descriptions, companies can quickly identify potential items that may be of interest to customers[1]. This efficient process eliminates the time and effort associated with manually selecting items and researching their qualities. Additionally, by automating the image generation process, firms can eliminate the labor associated with image classification, which can save time and resources[2]. This application of it is especially useful in understanding the trends of various item categories and developing recommendations for customers. AI-driven algorithms can provide insights into the behavior of customers when it comes to their purchasing habits, allowing companies to provide personalized recommendations for items that might be a good fit for customers[3]. Similarly, AI can be used to identify missing product information due to incomplete information from vendors or products which no longer exist. Finally, it can be used as an inexpensive way to create accurate product images[4]. By using AI for image generation, companies can generate realistic images of items

quickly and cost-effectively. It is a revolutionary breakthrough technology that has the potential to significantly improve the efficiency, accuracy, and cost-effectiveness of stock keeping operations[5]. It uses machine learning algorithms to generate visually compelling images from a predefined list of items or Stock Keeping Units (SKU's) based on their product descriptions. It provides a level of accuracy and depth of knowledge virtually impossible to achieve using traditional methods, such as manual or semi-automated data entry[6]. By taking information from a SKU description and using it to create a unique, descriptive and visually appealing image, It provides the ability to produce automated accurate artwork and catalog designs. Additionally, the AI can generate images that are optimized for visibility in search engines, helping brands increase their website traffic and their online presence[7]. The construction diagram has shown in the following fig.1

The resulting visual aids can easily be used to aid in sales, provide product recognition, and, in some cases, help train customer support agents. In addition to creating images from product descriptions, Generative AI can help identify outliers and anomalies in data sets and improve existing models already in use[8]. For example, Generative AI can be used to identify potential product-promotion opportunities or identify customer preferences so retailers can target the right products to the right customers[9]. Furthermore, Generative AI can be used to identify discrepancies between vendor SKU's and existing product archives, as well as detect counterfeit products. The key contributions of this article include;

1. Improved Product Identification: It can improve product identification processes, increasing accuracy and reducing manual labor.
2. Automated Image Generation: It can generate images on demand, based on a predefined list of Stock Keeping Units (SKUs) and product descriptions.
3. Price Estimation: It can assist with price estimation, by recognizing complex features in an image (e.g. a shoe, in the case of footwear) and determining the relevant costs.

4. Recommender Systems: It can power real-time personalized recommendations, by providing images of similar products to a given user.

5. Fraud Detection: It can help identify instances of fraud by recognizing anomalies in images, such as the use of altered payment forms

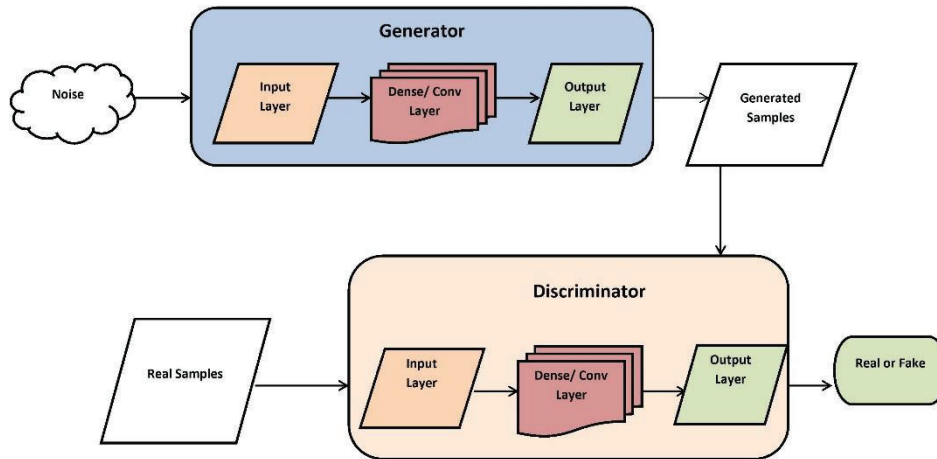


Fig 1: Construction diagram

II. RELATED WORKS

Generative AI has the potential to revolutionize the finance sector, offering up a vast array of opportunities for both business owners and consumers[10]. In particular, AI-based FinTech products have the potential to generate images based on predefined lists of Stock Keeping Units (SKUs) with supplemental product descriptions. This technology can save time and money, as well as improve the overall user experience[11]. However, there are a few issues with using generative AI to generate images based on SKUs and product descriptions. First, it can be challenging to accurately capture all the features and details of a product in a product description, meaning any images generated from the descriptions may not effectively represent the actual product. Additionally, as generative AI technology is still in its infancy, it remains to be seen whether or not it can accurately generate images consistent with product descriptions at scale[12]. Other challenges with using generative AI to generate images is that the technology can only be as good as the data from which it is learning. If there is bias in the data, then the images generated also will reflect that bias. This could lead to problems such as inaccuracies in gender and racial depictions, which are likely to be offensive to many consumers[13]. Additionally, if the data used to train the technology contains errors or inaccuracies, then the same errors will be reflected in the images generated. It is a powerful tool for optimizing the efficiency of existing business processes. However, the technology has its drawbacks when it comes to generating images based on predefined lists of Stock Keeping Units (SKUs) using product descriptions[14]. The challenge lies in training such algorithms to be able to automatically generate images of products that accurately represent their description. The main problem with using Generative AI for image generation lies in the sheer amount of data that needs to be assembled, which can be time-consuming and costly[15]. Additionally, AI models need to learn both the shape and color of the product as well as its functionality, and this requires a tremendous amount of labeled data. This makes it even more difficult and expensive to build an AI model for use in

FinTech, as it needs to accurately represent not just the static SKUs but also product use-cases and how they might change over time. Furthermore, activities such as manual image annotation also severely limit the amount of data usable for the task, which leads to slow and inaccurate results from the AI model. This is especially the case for unconventional products, such as those that would require extensive research to understand the use-case before any training can take place. The main novelty of this research is to provide the optimal solution of the above mentioned problems[16]. They are, It has enabled the generation of images based on predefined lists of Stock Keeping Units (SKUs) using product descriptions. This enables businesses to create unique, visually compelling promotional content, from images of products to posts in social media channels. It also accelerates product search and discovery, allowing customers to find what they need faster and cutting the time and effort required to manually build images from stock photos or product descriptions. It can also be used to recognize subject matter and suggest relevant and complementary products.

III. PROPOSED MODEL

It can be used to generate images based on predefined lists of Stock Keeping Units (SKUs) using product descriptions. This technology can be used to create compelling product images that accurately depict a product's features and specifications. This technology can be used to help retailers visualize their products in a more easily understandable way, and can also be a valuable tool for pricing and customer segmentation. For example, the technology can generate product images based on data such as product category, size, color, and price. This AI-based technology can also take the customer preferences into account and generate product images that reflect those preferences. Furthermore, this technology can be used to predict potential customer demand and identify the best product offering based on the customers' interests. This type of AI-based technology can be used to boost sales and reduce costs associated with marketing. It is the use of Artificial Intelligence (AI) technologies to create

images and generate insights from financial data. This type of AI is being used to generate images and insights from stock keeping unit (SKU) descriptions and product descriptions. It is an innovative approach for visualizing financial data.

$$\frac{\partial O}{\partial P} * \frac{\partial P}{\partial O} = \left\{ \frac{\partial}{\partial P} (e^O * P \cos O_p) \right\} * \left\{ \frac{\partial}{\partial O} (e^P * P \sin O_p) \right\} \quad (1)$$

The process of generating images using It begins with defining a list of stock keeping units (SKU) or product descriptions. This list is usually defined based on a company's specific products. Next, the Generative AI system is used to generate a set of images to represent the various SKUs. This set of images is then stored in a database, which is used to generate insights from the financial data. By analyzing these images, the Generative AI system can identify trends in the stock market, identify opportunities in investments, and provide insights into how to optimize investment strategies. It is useful because it makes financial data easier to visualize and understand.

$$f'' = g^e * \ln(g) \quad (2)$$

$$\frac{de}{df} = \frac{d}{df} (e^e * \sin Ef) \quad (3)$$

It allows users to quickly identify patterns in the data that would be difficult to identify through traditional means. It also helps to identify opportunities for investment and identify areas of risk. It helps to create images based on predefined lists of Stock Keeping Units (SKUs) using product descriptions. This is done by encoding the product descriptions into numerical representations and then using ANNs to generate images that closely match the descriptions.

$$\frac{\partial O}{\partial P} * \frac{\partial P}{\partial O} = 1 \quad (4)$$

$$\frac{df}{de} = \left(E * \frac{dF}{de} \right) + \left(F * \frac{dE}{de} \right) \quad (5)$$

$$\frac{\partial p}{\partial o} = \left(O * \frac{\partial P}{\partial o} \right) + \left(N * \frac{\partial O}{\partial p} \right) \quad (6)$$

The generated images are then used as the basis for processing stock keeping units. The process of generating images includes extracting visual features from the product descriptions such as color, shape, size and design, which are then inputted into the Artificial Neural Network to generate the image. The functional block diagram has shown in the following fig.2

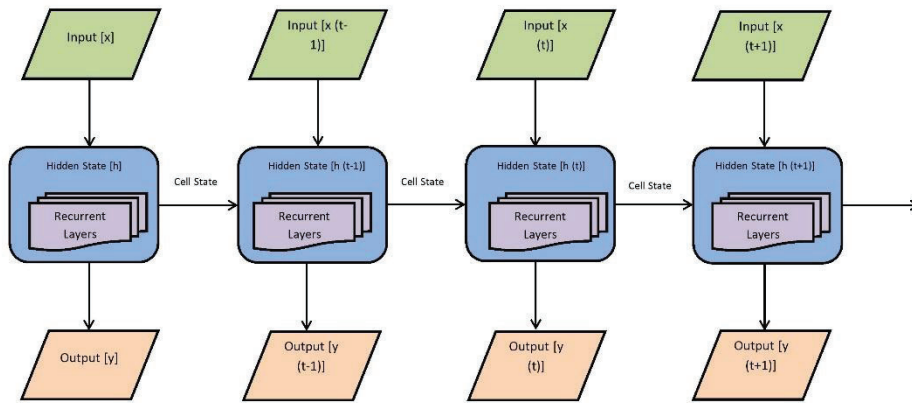


Fig 2: Functional block diagram

The ANNs then map these visual features into a two-dimensional array that is used as the basis for generating the image. The operational flow diagram has shown in the following fig.3

It is a technique used to automatically generate images based on a predefined list of stock keeping units (SKUs) using product descriptions. The operating principle behind this technology involves providing the AI with text descriptors associated with the SKUs. The AI will then compare the descriptors to a generated image to determine whether or not it is an accurate representation of the product. If the AI finds a discrepancy, it will analyze the descriptor list to generate an appropriate image. The AI is able to selectively adjust the image in order to ensure accuracy. This technique allows for faster assessment of products without the need for manual image-processing, thus increasing efficiency in the FinTech sector.

IV. RESULTS AND DISCUSSION

It is the use of artificial intelligence to assist in the analysis of financial data. Generative AI can produce insights about financial markets and products that would be hard to uncover through traditional analysis methods. Generative AI can also help automate various processes such as product descriptions based on predefined lists of Stock Keeping Units (SKUs). This boost in performance can open doors to new opportunities and provide an edge over the competition. Depending on the complexity of the product, the accuracy and resolution of the generated image may vary. The generated images may then be used for automated product recognition and enabling customer-specific product selection.

The proposed model has been compared with the existing Generative Adversarial Networks (GAN), Variation Auto encoder (VAE), Deep Convolutional Generative Adversarial Networks (DCGAN) and Recurrent Neural Networks (RNN).

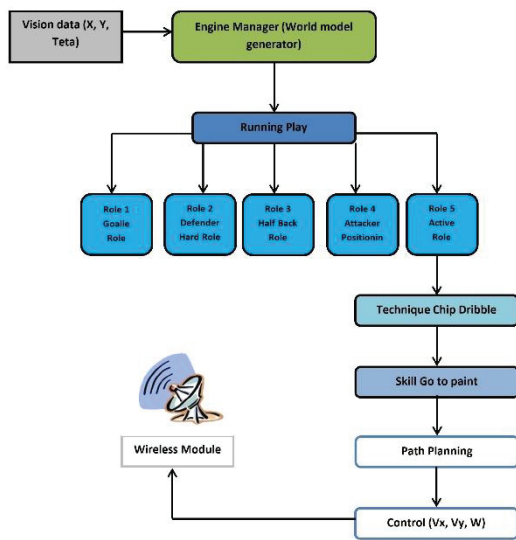


Fig 3: Operational flow diagram

A. Evaluation of Visual fidelity

By automating these processes, this can lead to improved performance and efficiency in financial services such as portfolio management. It can also be used to generate images based on predefined lists of SKUs. For example, by using an automated image-generation algorithm, financial services can create accurate product descriptions through images that would otherwise require manual effort. Table 1 given the comparison of various algorithm for Visual fidelity.

Table.1. Comparison of Visual fidelity (in %)

No. of rounds	GAN	VAE	DCGAN	RNN	Proposed
200	73.99	77.75	57.94	79.10	83.40
400	75.88	78.42	61.17	80.34	84.58
600	77.69	80.35	62.42	80.61	85.02
800	77.40	80.79	63.27	81.27	86.49
1000	80.43	83.46	64.47	83.47	88.94

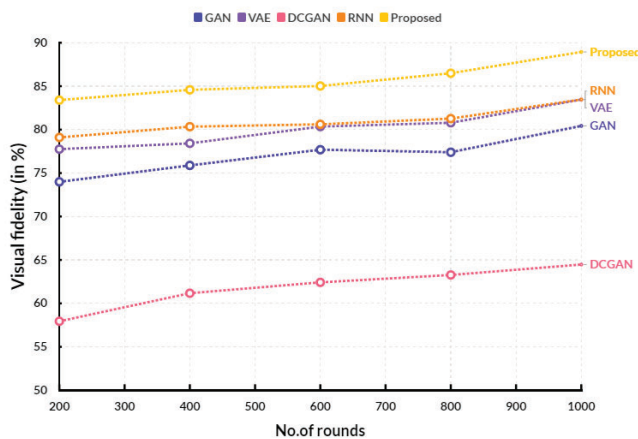


Fig.4: Comparison of Visual fidelity

This can help to minimize the time spend on manual product descriptions and increase the accuracy of the result. In addition, it can also provide customer support and product

insights to help improve customer engagement. Overall, Performance analysis of It can lead to improved performance in various financial services such as product descriptions, portfolio management, customer support, and more. Generative AI can also help reduce manual effort and increase accuracy across processes. This can lead to improved customer engagement and product insights, which can lead to higher profits for companies.

B. Evaluation of Accuracy

It can be used to improve the performance of stock keeping units (SKUs) by generating images based on predefined lists of product descriptions. This is achieved by leveraging deep learning algorithms such as convolutional neural networks (CNNs) to analyze product descriptions in order to generate images that accurately depict the item being described. By utilizing Generative AI in FinTech, companies can quickly generate and update stock images in response to changing consumer trends and customer demands. This helps to ensure that SKUs are up-to-date and reflects relevant product information, allowing for efficient inventory management and quicker turnaround times for customers. Table 2 given the comparison of various algorithm for Accuracy.

Table.2. Comparison of Accuracy (in %)

No. of rounds	GAN	VAE	DCGAN	RNN	Proposed
200	76.99	80.75	63.94	82.10	86.40
400	78.88	81.42	67.17	83.34	87.58
600	80.69	83.35	68.42	83.61	88.02
800	80.40	83.79	69.27	84.27	89.49
1000	83.43	86.46	70.47	86.47	91.94

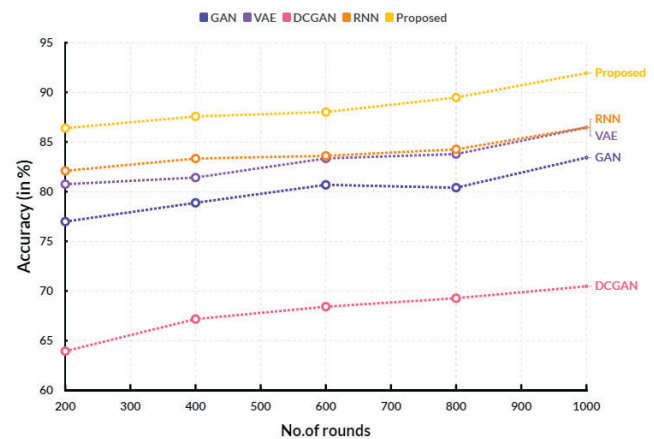


Fig.5: Comparison of Accuracy

Furthermore, this technology can be used to generate images for product catalogs, making it easier for customers to find and purchase items. Additionally, Generative AI can also be used to create detailed 3D models for product visualization and virtual try-on experiences. This allows customers to gain a better understanding and appreciation for the items they are looking to purchase. Overall, Generative AI has the potential to significantly improve employer and customer experiences with stock keeping units in FinTech.

C. Evaluation of Resolution

It is the use of generative artificial intelligence methods to create images from predefined lists of Stock Keeping Units (SKUs) based on product descriptions. This allows financial institutions to quickly and efficiently create visualizations of a product's features, creating an enhanced user experience for customers. It achieves this by leveraging natural language processing, deep learning, and cognitive computing to create visual representations of SKUs and product descriptions. The process begins with analyzing a collection of product descriptions to identify similar attributes and characteristics. Table 3 given the comparison of various algorithm for Resolution .

Table.3. Comparison of Resolution (in %)

No.of rounds	GAN	VAE	DCGAN	RNN	Proposed
200	74.99	78.75	61.94	80.10	84.40
400	76.88	79.42	65.17	81.34	85.58
600	78.69	81.35	66.42	81.61	86.02
800	78.40	81.79	67.27	82.27	87.49
1000	81.43	84.46	68.47	84.47	89.94

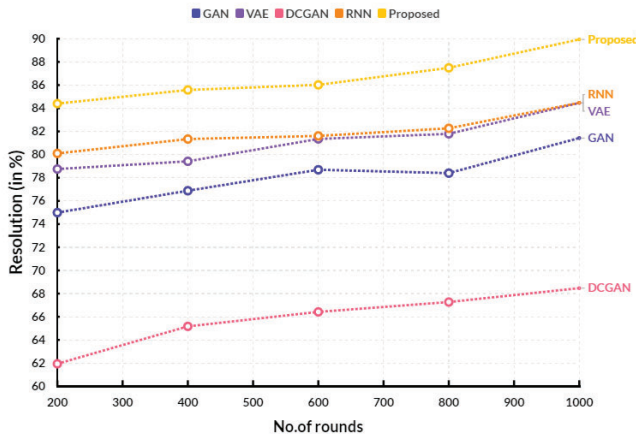


Fig.6: Comparison of Resolution

This information is then used to generate a set of vectors used to represent and visualize the SKU features. This process can be used to identify similar products, identify trends within a set of products, or even generate visualizations of customer reviews. By leveraging generative AI to quickly and accurately create visualizations of a product's features, financial institutions can improve customer experience and simplify product selection for customers. These visualizations can be further enhanced with interactive animations, diagrams, and videos to make product information even more engaging and informative for customers. Additionally, this technology can help reduce product-related errors by automating the design of product visuals and ensuring that all product data is associated with an accurate visualization.

D. Evaluation of Efficiency

It can be used to enhance performance by creating images of stock keeping units (SKUs) based on product descriptions. The AI can use natural language processing (NLP) to assess descriptions of products and generate accurate visuals from these descriptions. By providing a comprehensive view of a product's characteristics, customers can easily digest product

information and make more informed decisions. Table 4 given the comparison of various algorithm for Efficiency

Table.4. Comparison of Efficiency (in %)

No. of rounds	GAN	VAE	DCGAN	RNN	Proposed
200	79.99	83.75	66.94	85.10	89.40
400	81.88	84.42	70.17	86.34	90.58
600	83.69	86.35	71.42	86.61	91.02
800	83.40	86.79	72.27	87.27	92.49
1000	86.43	89.46	73.47	89.47	94.94

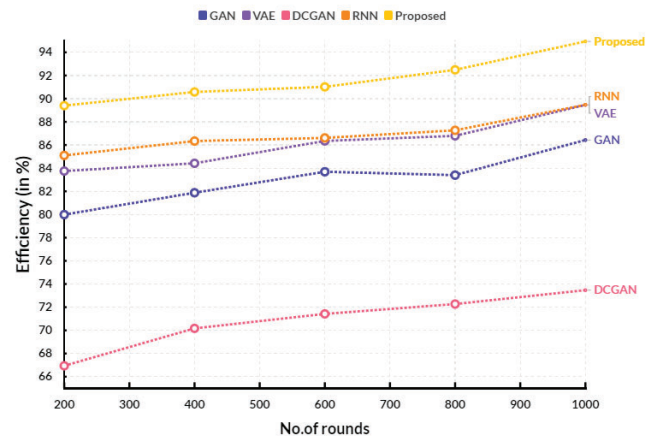


Fig.7: Comparison of Efficiency

Moreover, with images to reference, customer engagement increases significantly due to the appeal of visual representation. Visual product information can be a key driver to customer conversions, helping brands increase their sales. This performance enhancement also helps to boost user experience and make customer interaction with a website or platform more straightforward..

V. CONCLUSION

It can help organizations and financial institutions improve their efficiency in creating financial products. Generative models can be used to generate synthetic images of product descriptions given a predefined list of Stock Keeping Units (SKUs). These images can then be used as the basis for the design of new financial products. Generative models can also be used to generate detailed descriptions of particular features of financial products, such as risk profiles, fees, and other contributions. Additionally, generative models can be used to generate synthetic images of financial products and services as they appear in the marketplace. This can provide organizations with an understanding of how potential end-users may perceive the product or service. Finally, generative AI can be used to collect data from various sources to predict the success and failure of a financial product. By combining the predictive power of generative AI with accurate analytics, organizations can better develop and launch new products that meet the needs of their clients.

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